

Shreyas M Kaushik

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SUMMARY

Result-oriented Senior Software Engineer specializing in AI/ML systems and workflow orchestration, with 3+ years of experience building scalable applications from scratch and owning end-to-end systems. Proven track record in architecting and implementing cloud-native solutions that process millions of requests daily using Netflix Conductor and modern microservices architecture. Strong expertise in Python and TypeScript with a focus on FastAPI and Node.js. Demonstrated leadership through mentoring junior developers and driving technical transformation initiatives. Passionate about solving complex problems with pragmatic engineering solutions.

EXPERIENCE

- IHX - A Perfios Company** Bangalore, India
Senior Software Engineer 01-07-2024 - Present
 - Netflix Conductor Orchestration:** Architected and implemented comprehensive workflow orchestration using Netflix Conductor for document processing pipelines, managing complex extraction, OCR, classification, and quality control tasks. Designed dynamic workflow definitions with parallel processing patterns, reducing processing time by 40% and improving system reliability.
 - Persistence Engine:** Architected and developed a single-layer persistence solution using TypeORM and Node.js/TypeScript that reduced average query time from 1-2 seconds to 10-15ms across 10 tables, significantly improving system performance and scalability.
 - Status Notification Service:** Built a high-throughput system processing over 10 million messages daily using SNS to manage and track processes across various AI/ML entities, providing unified status visibility to clients through real-time dashboards.
 - Context-Aware Service:** Built a FastAPI service for Bill Line Item Classification that selects appropriate models or rule-based logic based on insurance and policy combinations, optimizing claim processing efficiency and accuracy by 25%.
- IHX - A Perfios Company** Bangalore, India
Software Engineer 16-08-2022 - 30-06-2024
 - Document Intelligence Platform:** Developed a comprehensive document processing solution for India's largest health benefits administrator that increased "complete claim submissions on first instance" from 23% to 77% within 2 weeks by providing secure integration to classification models and intelligent routing.
 - Infrastructure Transformation:** Led infrastructure design and implementation for securing AI models and establishing efficient scaling strategies using Kubernetes and AWS services. Introduced comprehensive monitoring with Grafana, Loki, and Prometheus, enabling rapid deployment of new features.
 - Quality Control System:** Created React-based Quality Control UI screens with real-time collaboration features for correction of ML model outputs, contributing to an increase in document classification model accuracy from 64% to 96% through improved human-in-the-loop workflows.
 - Authentication Solution:** Designed and implemented a comprehensive MFA and SSO solution using Keycloak with custom plugins, enabling secure access control to multiple services and applications while reducing login time by 50%.

TECHNICAL SKILLS

- Languages:** TypeScript, Python, JavaScript, Bash, SQL
- Web Technologies:** Node.js, React.js, Django, FastAPI, Express, RESTful APIs
- Workflow Orchestration:** Netflix Conductor, Apache Airflow, SQS, Event-driven architecture
- AI/ML Integration:** Model deployment, API wrapping, Classification pipelines, Computer Vision
- Cloud & DevOps:** AWS (EC2, S3, SQS, ALB, EKS, API Gateway, Cognito), Docker, Kubernetes, GitHub Actions
- Databases:** MongoDB, PostgreSQL, MySQL, TypeORM, Mongoose
- Monitoring & Security:** Grafana, Prometheus, Promtail, Loki, Keycloak, OAuth 2.0, JWT, MFA

PROJECTS

- AI Video Ad Stitching Service (Python, OpenCV, ML):** Developed a video processing service using computer vision and machine learning for automated ad insertion and content stitching. Implemented real-time processing pipeline for content recognition and seamless video transitions.
- Secure Files Demo (Django, React, Keycloak):** Created a comprehensive demo application showcasing MFA plugin integration with additional security practices like self-signed certificates and role-based access control.  github.com/shreyasY2k/secure-files
- Contra (Three.js, AI-assisted):** Built a complete 3D run & gun shooter game in just 2 hours leveraging AI tools like GitHub Copilot and Claude. Features advanced enemy AI, vehicle physics, and weapon systems with particle effects.  shreyasy2k.github.io/contra
- Python Spawn (Node.js, npm package):** Created and published an npm package that bridges Node.js and Python with 500+ weekly downloads, enabling seamless execution of Python scripts and functions from Node.js applications. Features automatic Python environment detection and robust error handling.  github.com/shreyasY2k/python-spawn

EDUCATION

- **Sapthagiri College of Engineering** Bangalore, India
Bachelor of Engineering - Information Science; CGPA: 8.5 *2018 - 2022*

TECHNICAL WRITING

- **Building a Custom MFA Plugin for Keycloak:** 🔗 SMS, Email & Telegram Integration
- **Mastering Multi-Factor Authentication in Keycloak:** 🔗 SMS, Email, and TOTP Setup Guide
- **Unlocking Seamless Access:** 🔗 Orchestrating Centralized User Management with Keycloak

CERTIFICATIONS

- **Architecting with Compute Engine:** 🔗 Coursera Specialization Certificate
- **The Modern Javascript Bootcamp:** 🔗 Udemy Certificate